Points to check before an unmanned system is used on ground suspected of containing unexploded ordnance

The following points need to be settled, whichever supplier is involved, before an unmanned system is used on ground suspected of containing unexploded ordnance. They are meant as a working basis for assessing offers and trials.

Area	What to check	Basis
Operating category	whether the operation stays in the open category A3. Dropping objects and carrying dangerous goods are not permitted there.	Regulation (EU) 2019/947
Exemption for public safety authorities (BOS)	whether the exemption under § 21k LuftVO applies. It has to be checked before every deployment, and it also covers third parties flying on behalf of the BOS.	§ 21k LuftVO, EGRED 2
Minimum standards	whether the operator meets the national minimum standards for drone operations by or on behalf of the BOS.	EGRED 2 (BBK, 2nd edition 2023), STS-BOS (LBA, 16 July 2025)
Qualification and crew	Proof of the remote pilots' qualifications. A2 or equivalent internal training is recommended, together with a minimum crew for each deployment.	DFV and AGBF, Fachempfehlung Drohnen und UAV, August 2025
Airspace	Coordination with crewed aviation, in particular police helicopters and air tankers, and visibility via ADS-B or FLARM.	the same recommendation
Extinguishing agent, product law	whether the agent may be placed on the market in Germany and the EU. CE marking, conformity, which state market surveillance authority is responsible.	ProdSG
Extinguishing agent, explosives law	whether a permit is required, whether it is classed as a pyrotechnic article, how storage and transport are regulated.	§ 7 SprengG, 1. SprengV
Failed drop	what happens to a payload that has not gone off, on ground that nobody is allowed to enter.	Deployment planning
Ordnance disposal	Involvement of the KMBD, which decides what is possible on the ground.	State law (Landesrecht)
Operating limits	Wind, thermals above the fire, smoke, behaviour on loss of link, fallback modes.	Manufacturer data, evidence
Liability	Sum insured, co-insurance of the authority, risk of a secondary fire from a failed drop.	Contract

The points on product law and explosives law are the ones most often missing in conversations. They apply to every supplier of a system that drops something, including us.

Disclosure: Averium Dynamics develops firefighting drones. The points above apply to us exactly as they do to any other supplier, and there are two of them we cannot yet fully answer.

From the article “Munitions-contaminated land: the number everyone quotes is not what the directive says”, with sources: averiumdynamics.com/blog/munitions-contaminated-land-what-the-rules-say

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